



Chapter 1

Introduction



Nội dung

- 1.1 Giới thiệu
- 1.2 Các giải thuật trong học máy
- 1.3 Xây dựng hệ thống học máy
- 1.4 Tập dữ liệu chuẩn



What Is Machine Learning?

[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed.

—Arthur Samuel, 1959

A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

—Tom Mitchell, 1997

Machine Learning is the science (and art) of programming computers so they can *learn from data*.



Why Use Machine Learning?

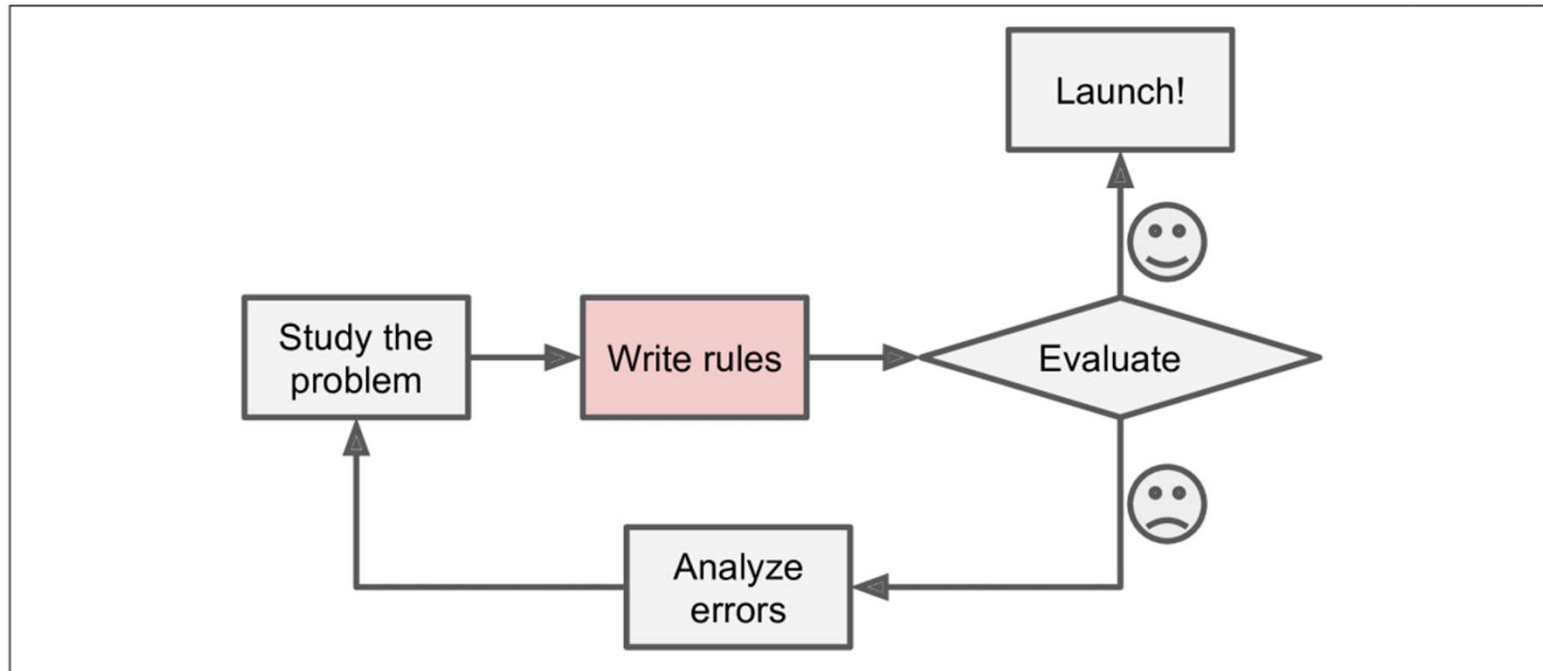


Figure 1-1. The traditional approach



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Why Use Machine Learning?

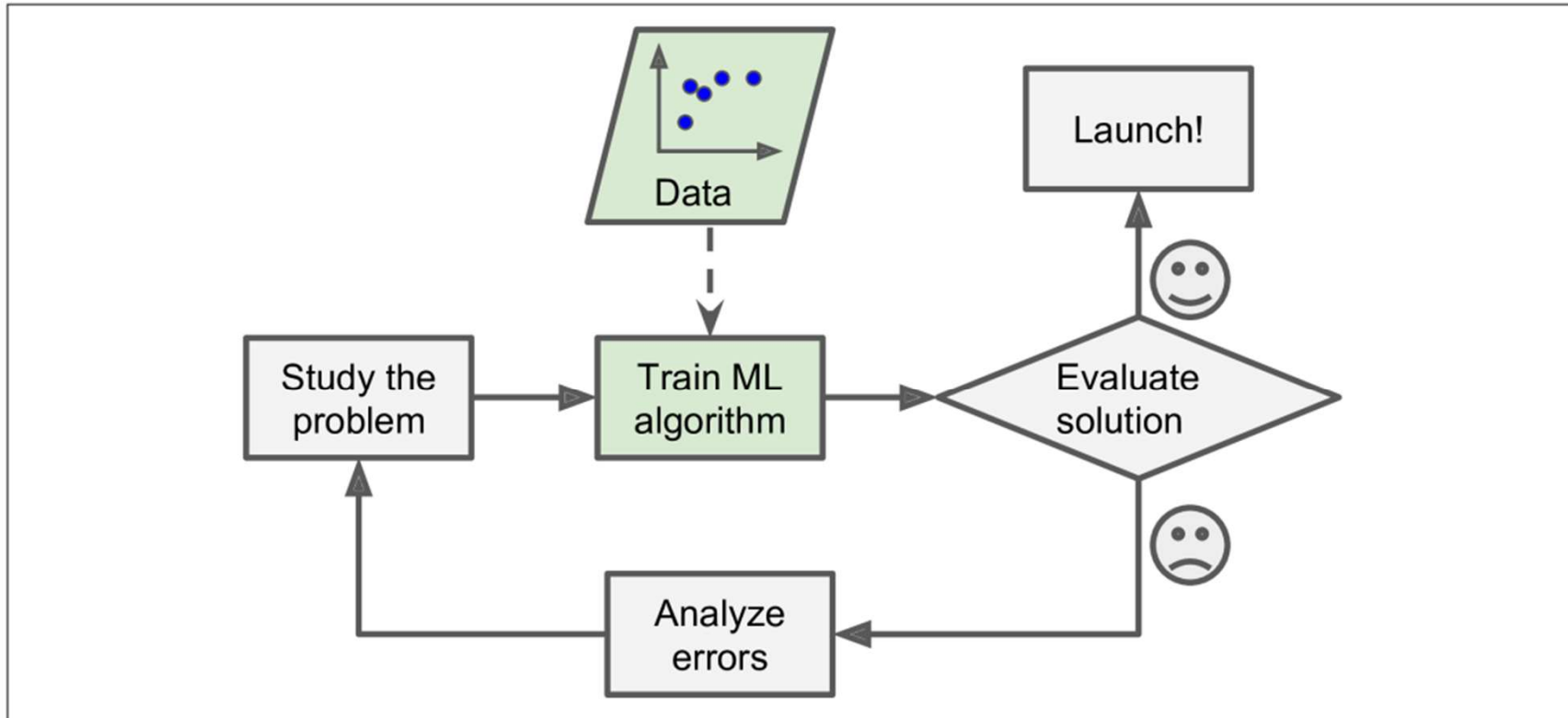


Figure 1-2. Machine Learning approach



Why Use Machine Learning?

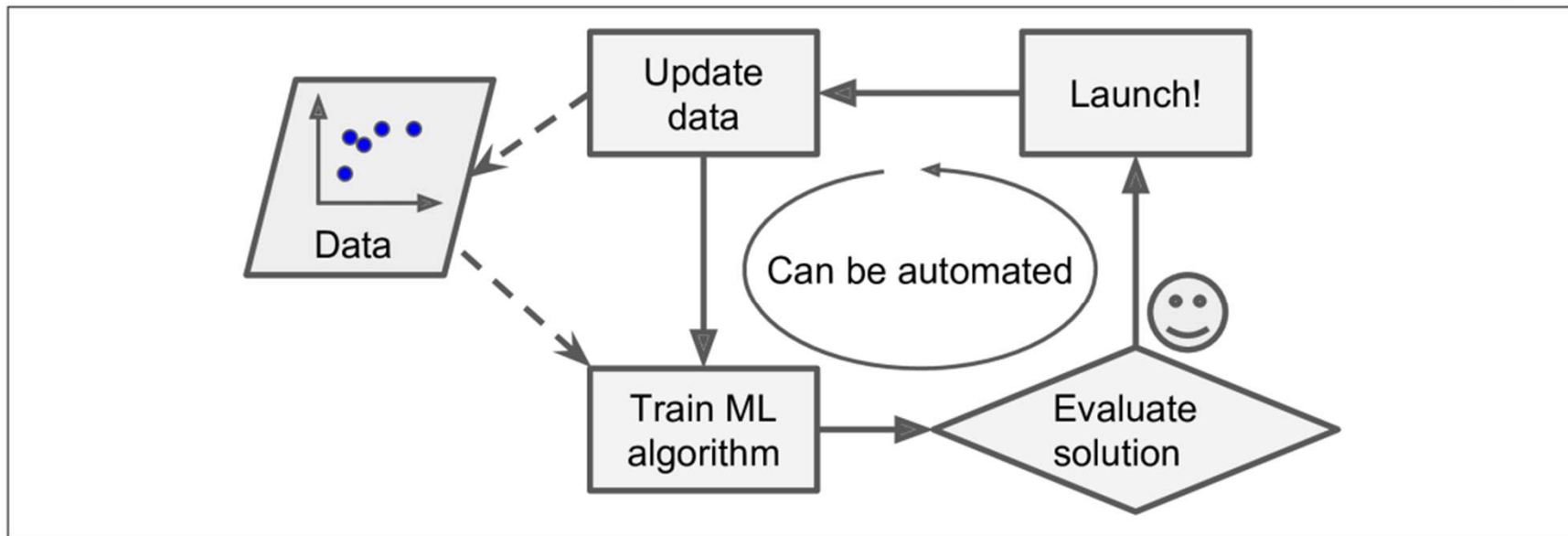


Figure 1-3. Automatically adapting to change



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Why Use Machine Learning?

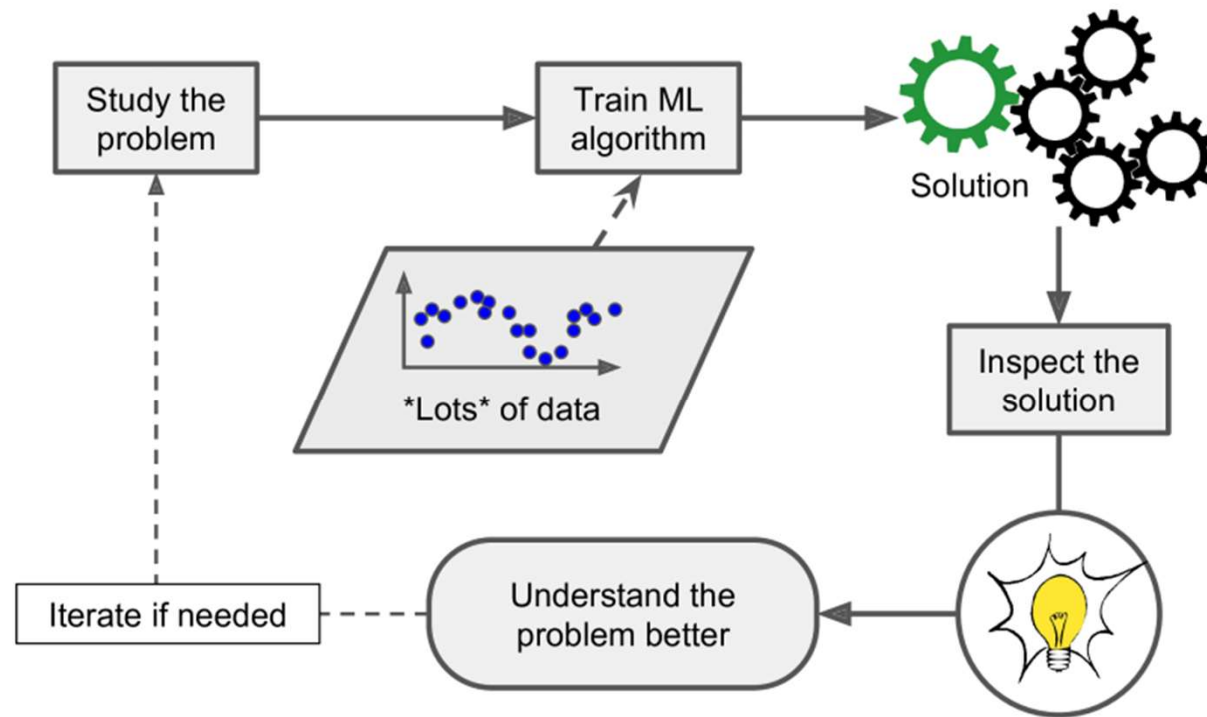
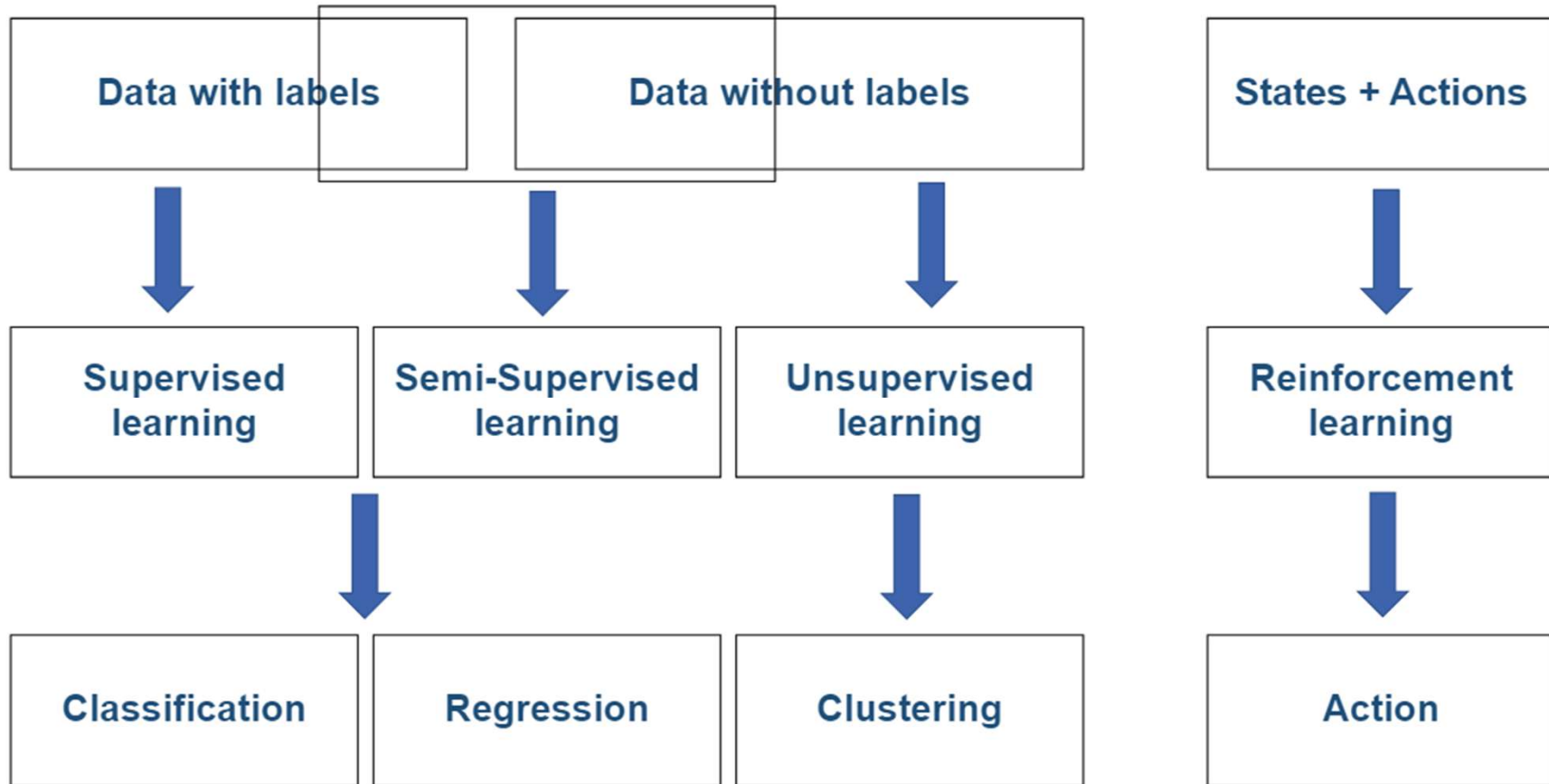


Figure 1-4. Machine Learning can help humans learn



Types of Machine Learning Systems



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Supervised learning

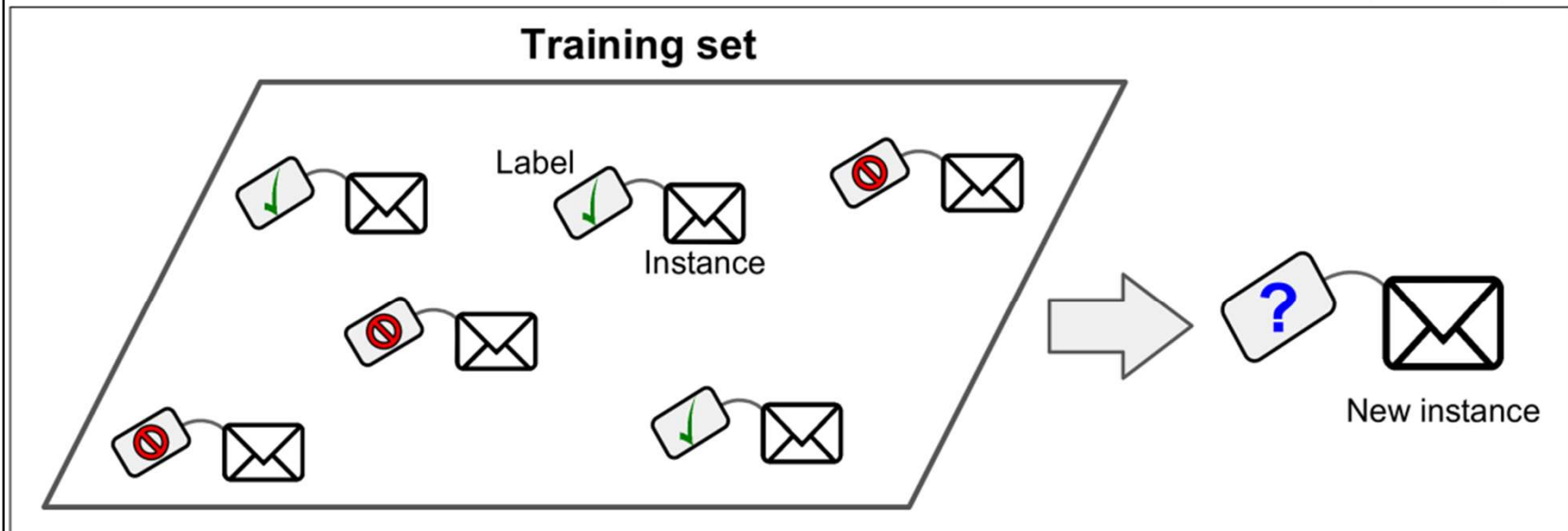


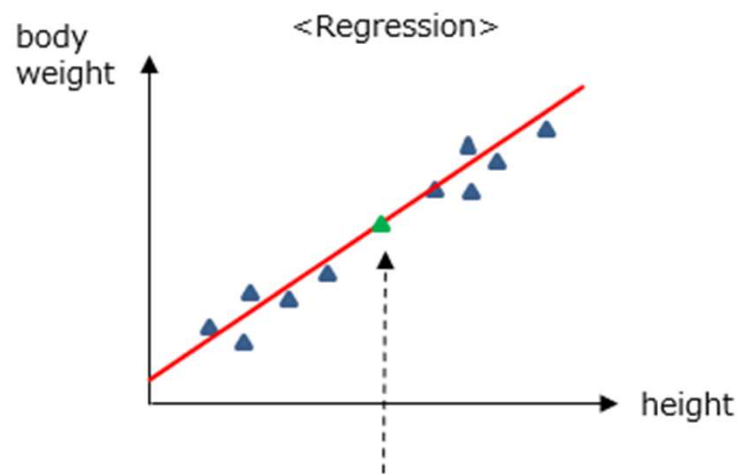
Figure 1-5. A labeled training set for supervised learning (e.g., spam classification)

Supervised learning

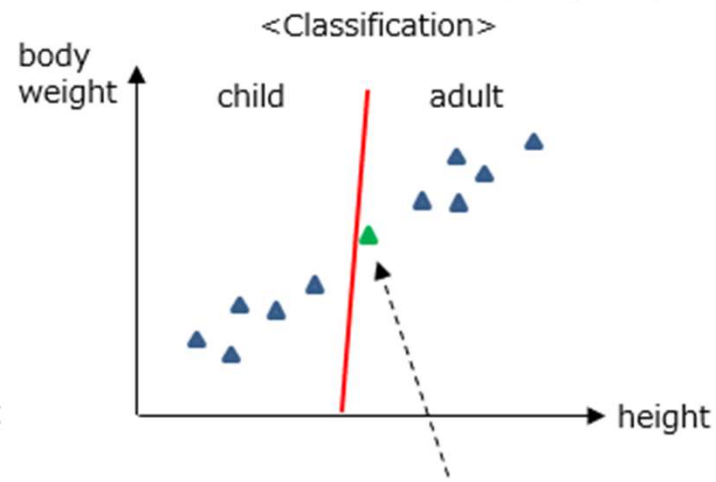


Figure 1-6. Regression

Supervised learning



Predict the characteristics of new data "numerically"(weight value)



Predict the characteristics of new data by "attributes" (adults)



- Here are some of the most important supervised learning algorithms:
 - k-Nearest Neighbors
 - Linear Regression
 - Logistic Regression
 - Support Vector Machines (SVMs)
 - Decision Trees and Random Forests
 - **Neural networks**

Unsupervised learning

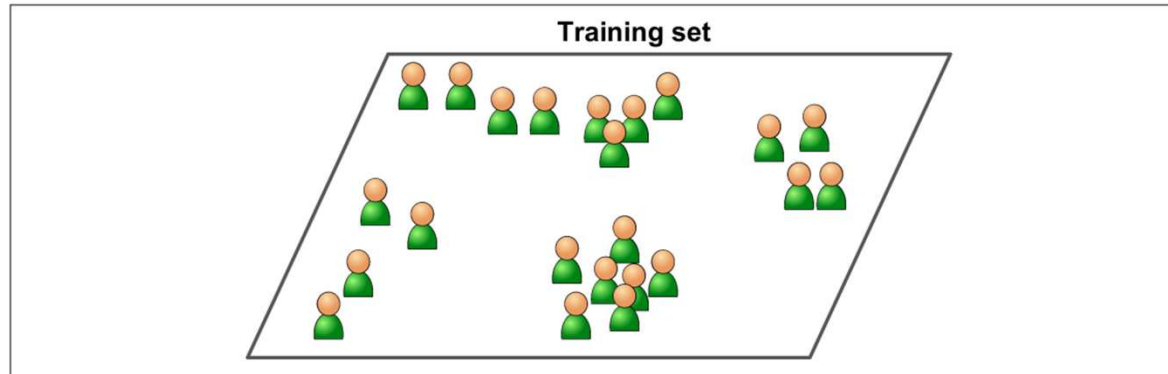


Figure 1-7. An unlabeled training set for unsupervised learning

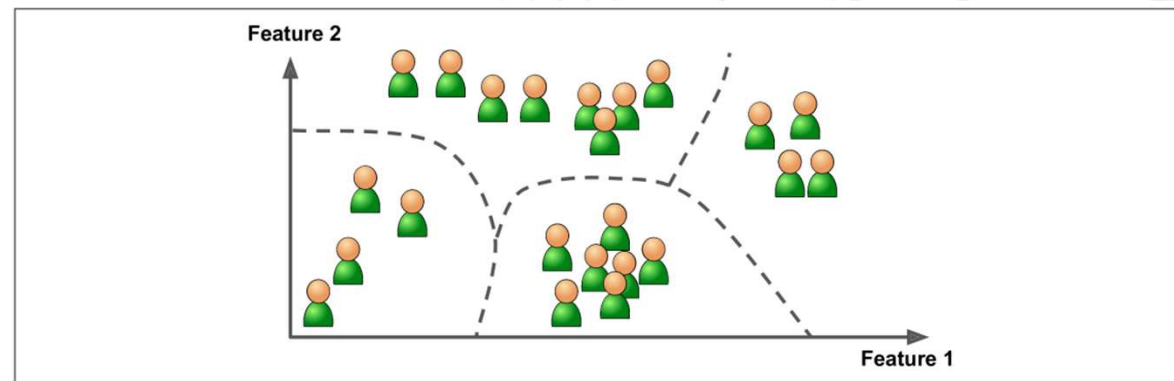


Figure 1-8. Clustering

Unsupervised learning

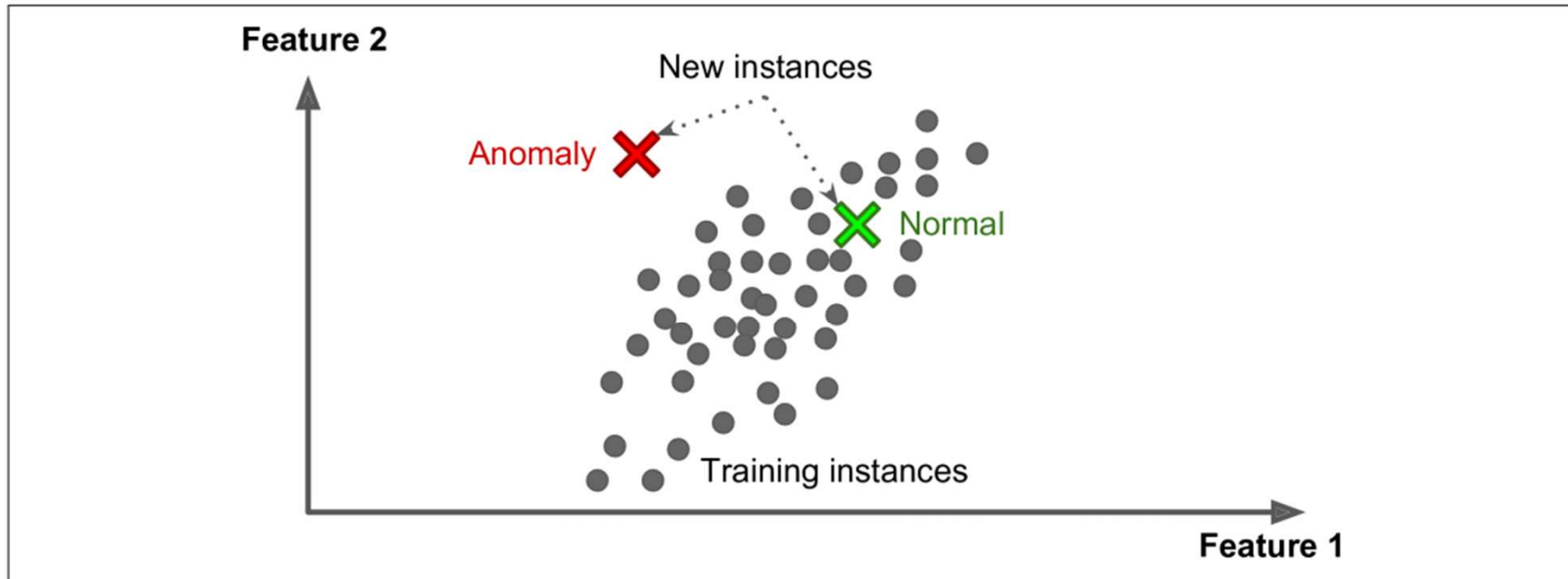
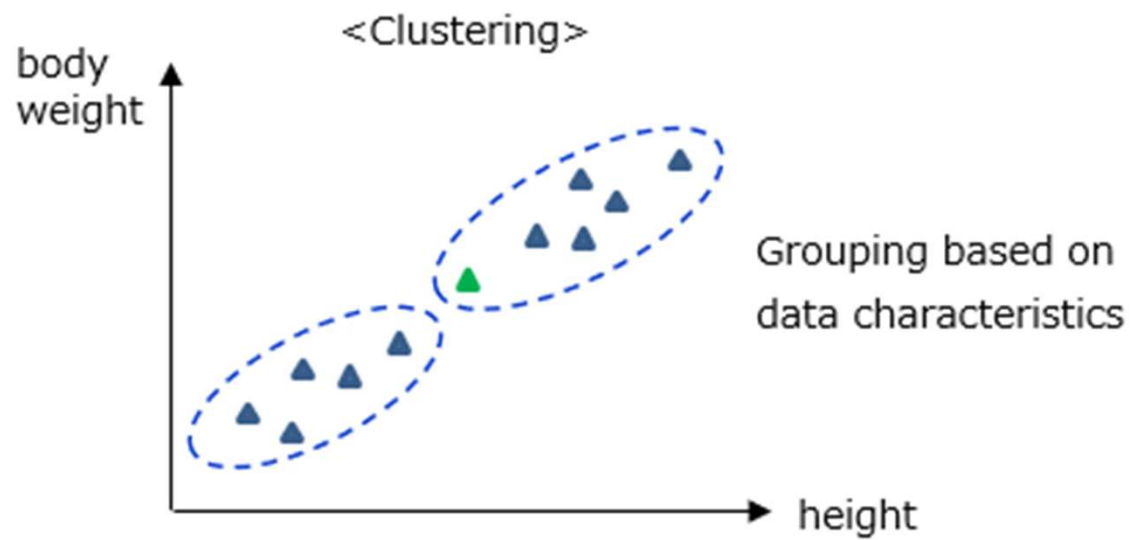


Figure 1-10. Anomaly detection

Unsupervised learning





Here are some of the most important unsupervised learning algorithms:

- Clustering
 - K-Means
 - DBSCAN
 - Hierarchical Cluster Analysis (HCA)
- Anomaly detection and novelty detection
 - One-class SVM
 - Isolation Forest

Neural networks *

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Semisupervised learning

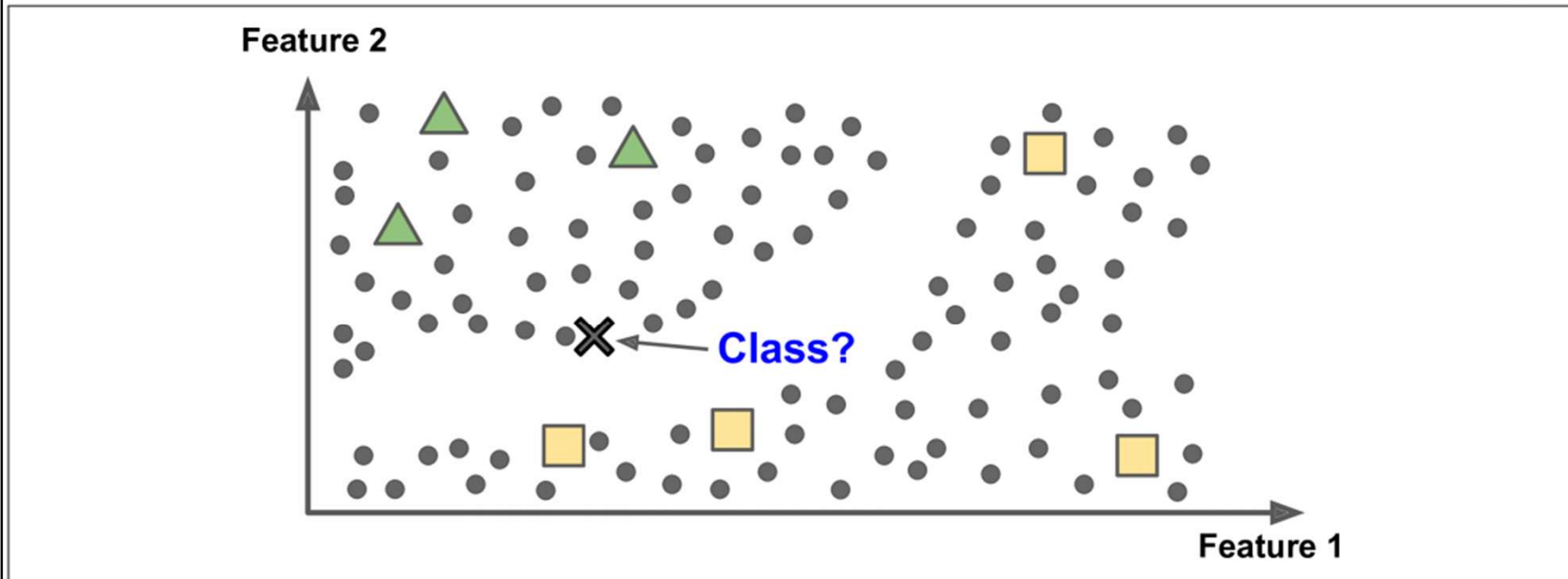


Figure 1-11. Semisupervised learning

Reinforcement Learning

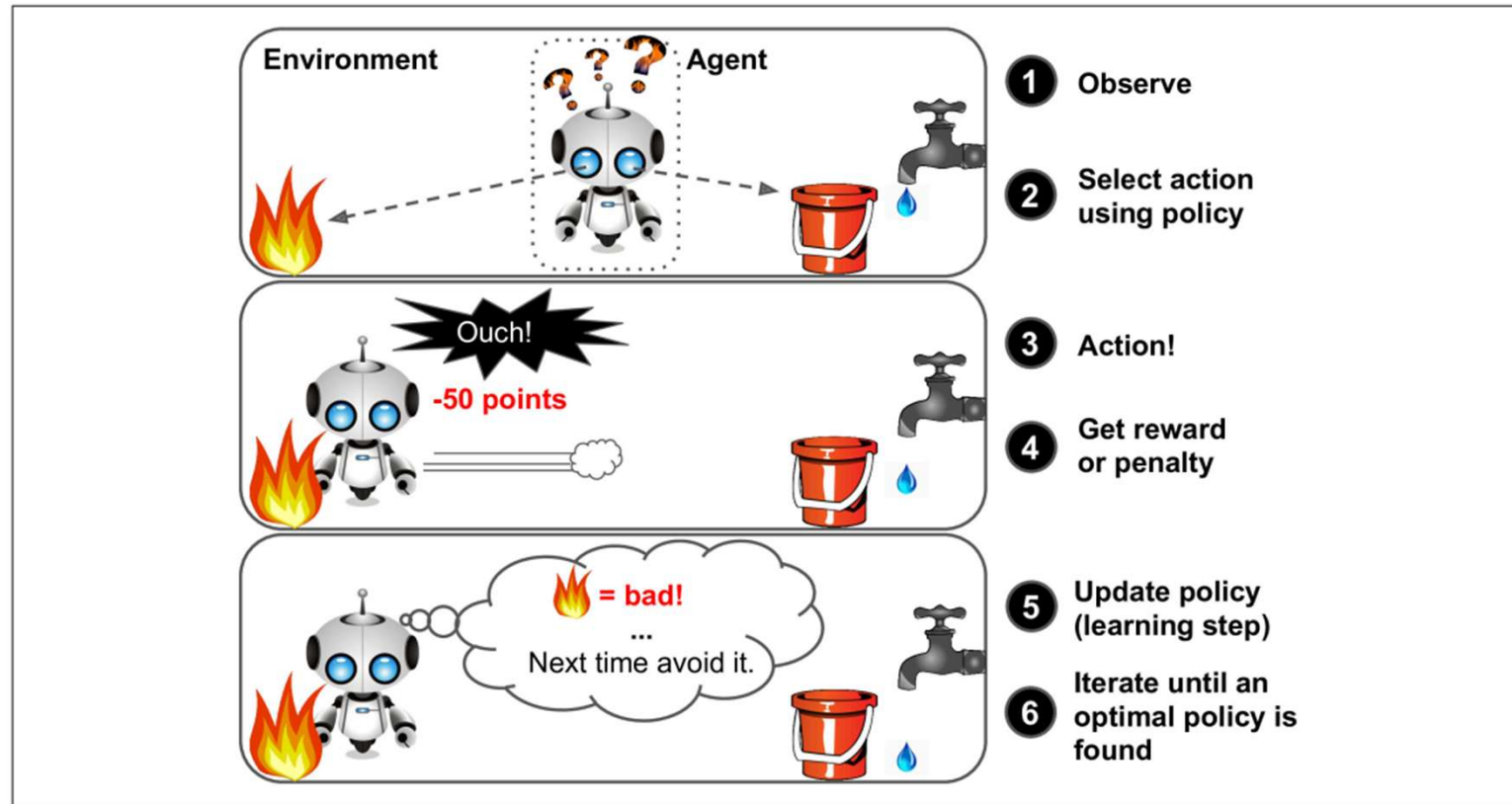


Figure 1-12. Reinforcement Learning

Batch and Online Learning

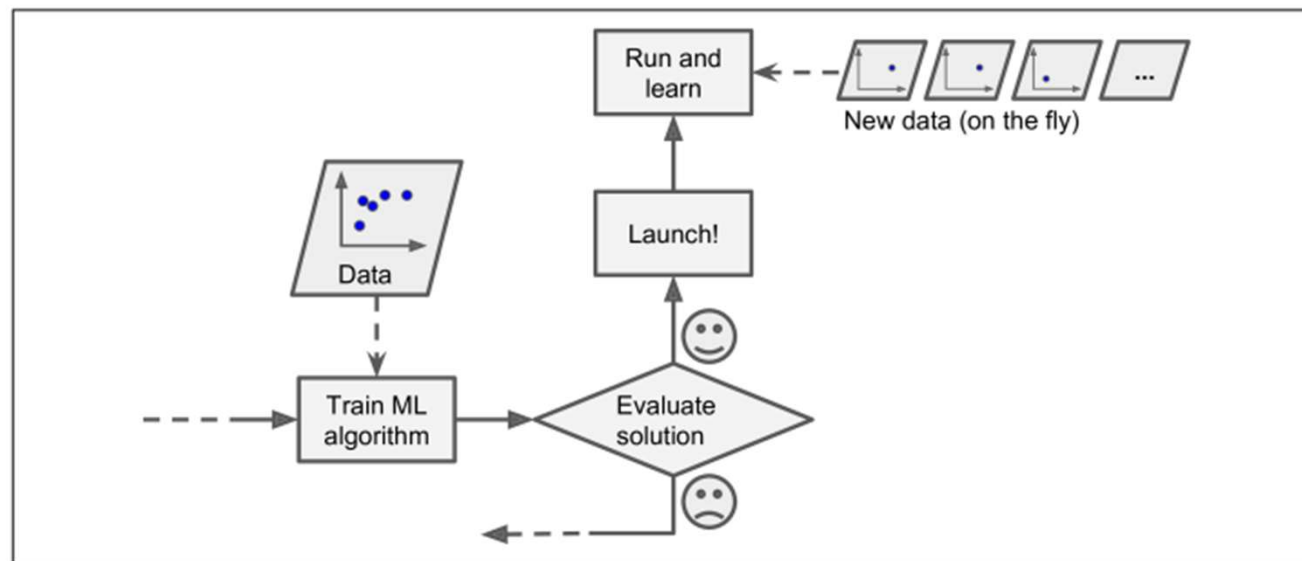


Figure 1-13. Online learning

Batch and Online Learning

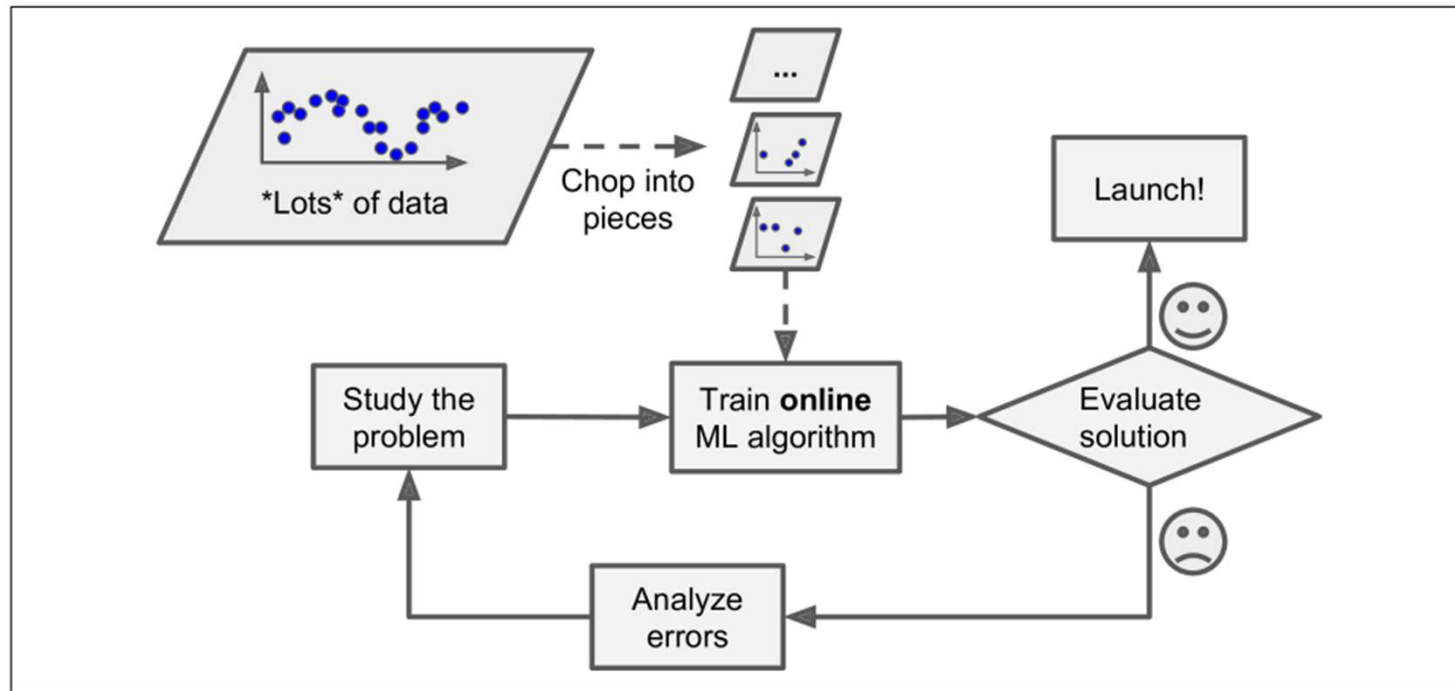


Figure 1-14. Using online learning to handle huge datasets



Instance-Based Versus Model-Based Learning

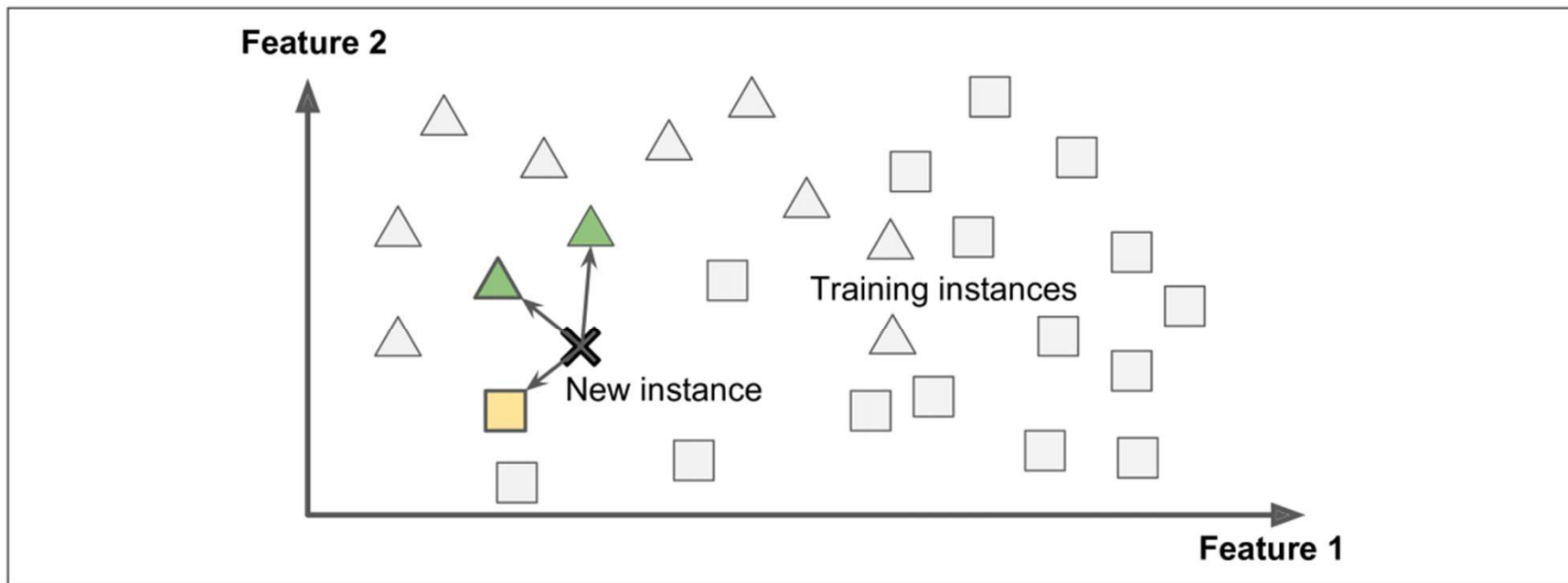


Figure 1-15. Instance-based learning



Instance-Based Versus Model-Based Learning

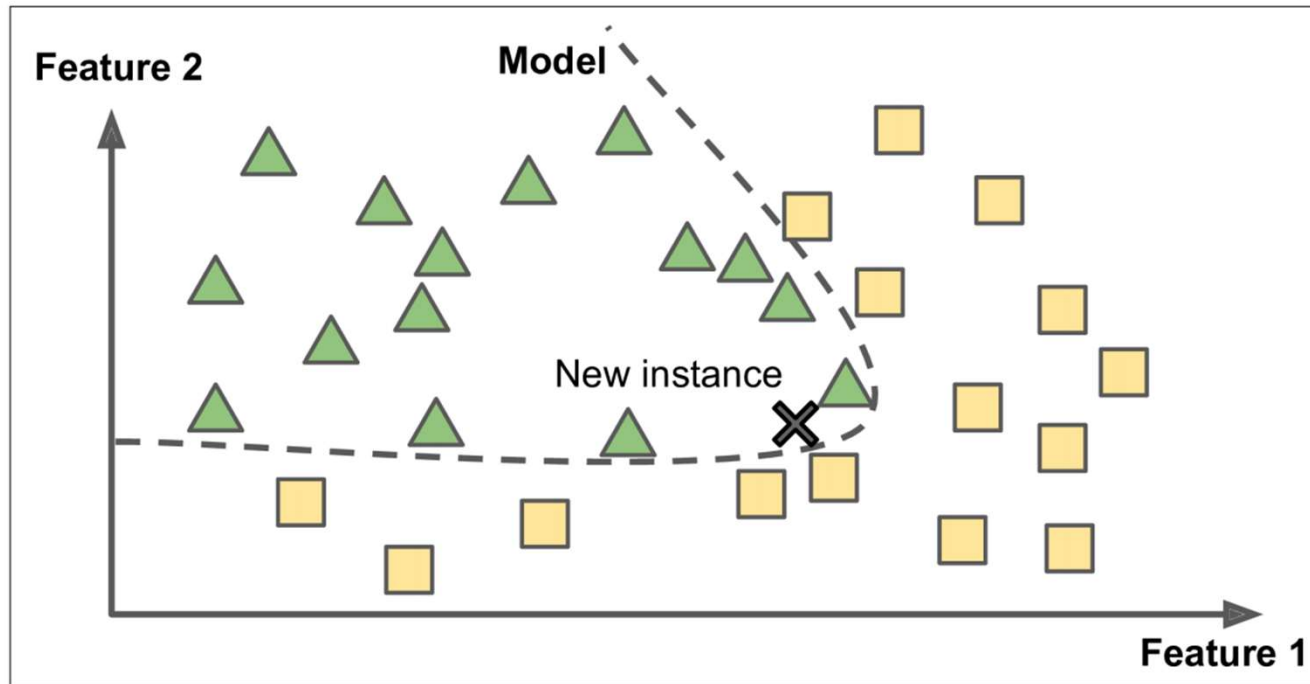


Figure 1-16. Model-based learning



Main Challenges of Machine Learning

- Insufficient Quantity of Training Data
- The Unreasonable Effectiveness of Data
- Nonrepresentative Training Data
- Poor-Quality Data
- Overfitting the Training Data
- Underfitting the Training Data
- Testing and Validating
- Hyperparameter Tuning and Model Selection

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Thank you!

ĐẠI HỌC CÔNG NGHIỆP THÀNH PHỐ HỒ CHÍ MINH

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